TECHNICAL NOTE

Governance costing model: methodological note (v018)

Emanuel Marques Queiroga · 9 November 2026 · AI in Education Procurement Tool · v018

Summary

The governance cost is calculated with a parametric costing model, not an econometric one: no value is estimated by regression. Each option chosen in the plan becomes an explicit calculation, with parameters that the user can see and change in the tool.

Most of the cost comes from people. For 25 options, the tool uses a reference team: the roles needed, how many people, each person's time share and for how many months. The monthly cost of each role is the country's average earnings for that occupational profile, published by the ILO (ILOSTAT), plus the country's employer social security contribution (SSA and ISSA).

Three additional costs are added on top of the options, each with a percentage from a public source: travel and in-person activities (25%), overhead (15%) and administrative fee (7%). Three questions determine whether each one applies.

Each number carries an evidence grade: A when it comes from the country's own data, B when it comes from a public source or from the median of similar countries, and C when it is a tool convention still to be calibrated. The composition of the teams is grade C and is the part of the model that most needs validation.

Scope and cost structure

Governance has 3 challenges, 9 categories and 32 options; each option has a fixed calculation, defined by its cost type. Categories 2.1 and 3.2 have no options: the category itself is the cost line.

Cost type	Where it applies	Calculation	Source of the unit value
Reference team	25 options: categories 1.1, 1.2, 1.3, 2.1, 2.3 and 3.3	Sum over roles: people × time share × months × monthly cost of the profile	Country earnings by profile (ILOSTAT) × country employer contributions
Full-time staff	4 options in category 3.1	People × annual cost × months ÷ 12	Country earnings, profile 2, × contributions × 12; for "IT experts", a reference annual cost of US$45,000
Training	Category 3.2	People × hours × cost per hour	US$12 per participant-hour
Entered amount	4 options in category 2.2 (cloud, hosting, storage, computing)	Total amount entered by the user	No reference: shown as "missing amount" until entered
Percentage	Additional costs: travel, overhead, administrative fee	Percentage × base	Global Fund, 2 CFR 200.414, UNSDG policy

The four infrastructure options have no price reference because their cost depends on the provider, the data volume and the procurement model. In category 3.1 and in training, the number of people and hours is always entered by the user.

Notation and general equation

The total cost of the governance plan is the sum of the cost of each selected option, plus three additional costs calculated on that sum.

$$C = D + V + O + T, \qquad D = \sum_{o \in \mathcal{O}} C_o$$

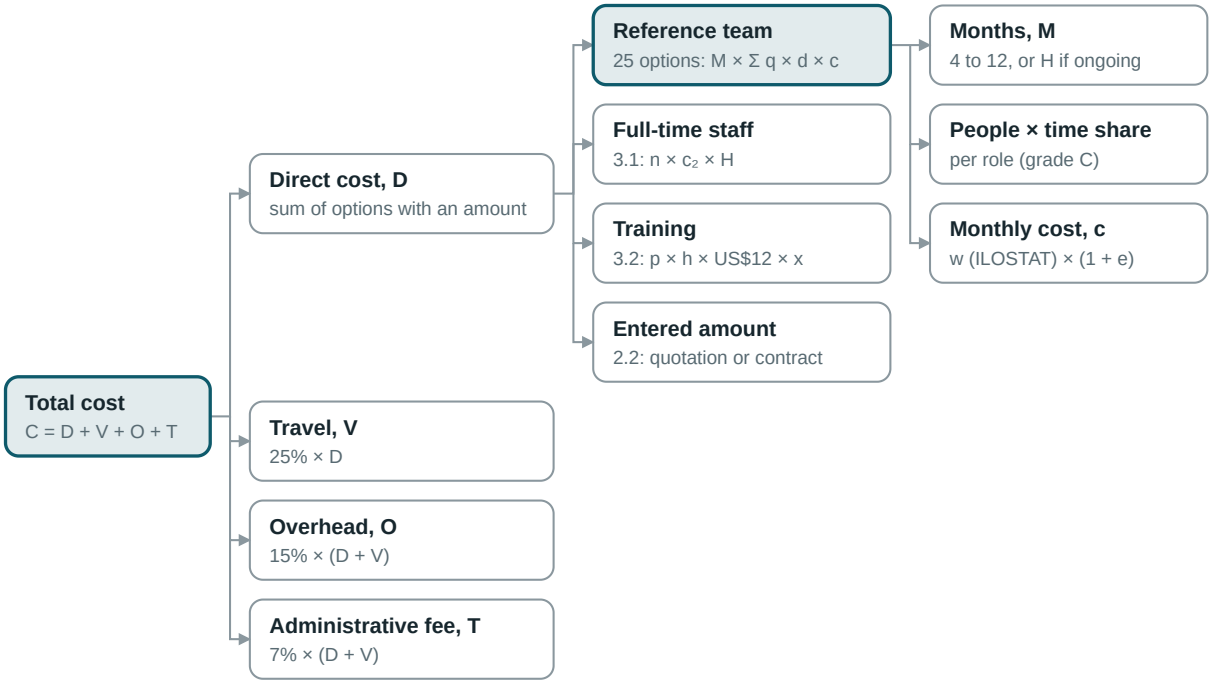


Figure 1. Structure of the governance cost, from the total to the parameters.

Uncertainty is concentrated in the reference-team branch: people, time shares and months are tool conventions, while earnings, contributions and percentages have public sources.

Symbol	Meaning	Unit
C	Total cost of the governance plan	plan currency
D	Direct cost: sum of the options that already have an amount	plan currency
C_o	Cost of option o	plan currency
V, O, T	Travel and in-person activities, overhead, administrative fee	plan currency
H	Plan duration (default: 12)	months
M_o	Working months of option o (one-off) or H (ongoing)	months
q_r, d_r	People and time share of role r in the reference team	people, share of time
w_g	Average monthly earnings of profile g (ISCO-08, groups 1 to 4) in the country	plan currency per month
e	Country employer contribution on payroll	share

Symbol	Meaning	Unit
x	Exchange rate: plan currency units per US$	plan currency per US$

An option without an amount (for example, infrastructure with no price entered) stays out of D and is shown as "missing amount". The additional costs are therefore calculated only on what already has an amount and grow as pending options are priced.

Options with a reference team

The cost of an option with a reference team is the team's monthly cost times the working months; the monthly cost adds up, role by role, people × time share × monthly cost of the profile.

$$C_o = M_o \sum_{r \in R_o} q_r \, d_r \, c_{g(r)}, \qquad c_g = w_g \, (1 + e)$$

Each role r has a profile g(r), one of the four ISCO-08 major groups: 1, managers (coordination, unit heads, technical leads); 2, professionals (specialists, data scientists, developers); 3, technicians and associate professionals (technical analysts, school liaisons); 4, clerical support workers.

Working months

A one-off option (a regulation, a guideline, a sandbox) has its own reference months, from 4 to 12. An ongoing option (supervision, committees, coordination, data management, open-source maintenance, institutional unit) lasts the whole plan, so M_o = H.

Country earnings

The earnings w_g are the monthly average of employees in group g, from the ILOSTAT indicator "Average monthly earnings of employees by sex, occupation and currency", both sexes, latest year available since 2015 (data fixed in the tool on 9 November 2026). When the country has data (grade A), the tool uses the local-currency value if the plan is in the country's currency; otherwise it converts the ILO's own US$ value with the exchange rate x.

For countries without data, w_g becomes the US$ median of countries with data in the same World Bank region and income group, with at least 3 countries, or, failing that, the median of the income group (grade B). Values below US$30 a month are discarded as recording errors.

Employer contributions

The contribution factor 1 + e uses the employer social security contribution, as a percentage of payroll, from table 4 of Social Security Programs Throughout the World (SSA and ISSA; Africa and the Americas, 2019; Europe and Asia and the Pacific, 2018). Without contribution data, e = 0. In countries with a contribution ceiling, or where employers pay benefits directly, the factor understates the actual cost.

When the country has neither earnings nor an estimate ("other" country), the monthly cost of every role is the generic loaded-cost reference, US$36,000 a year ÷ 12 × x, which already includes contributions (grade C).

Editable values

In the tool, the user can change the months of a one-off option, the number of people in each role and the monthly cost of each role. A changed value replaces the reference, and the line moves from "estimated" to "entered" once all its values have been entered.

Other options: staff, training and infrastructure

Options without a reference team use simpler calculations, in which the quantity is always entered by the user and only the unit value may come from a reference.

Full-time staff (category 3.1)

The options "Legal professionals", "AI governance personnel" and "AI and education specialists" multiply the number of full-time people n, entered by the user, by the country's annual cost for profile 2 and by the share of the year covered by the plan.

$$C_o = n \cdot 12\, w_2(1+e) \cdot \frac{H}{12} = n\, w_2(1+e)\, H$$

The option "IT experts (AI, data, security, infrastructure)" uses, instead of country earnings, the reference annual loaded cost for integration and infrastructure engineering, US$45,000 × x (grade C), with no additional contributions.

Training and professional development (category 3.2)

The cost of staff training is the number of participants p times the hours per participant h times the cost of a training hour, whose reference is US$12 per participant-hour (grade C).

$$C_o = p \cdot h \cdot 12\, x$$

Digital infrastructure and hosting (category 2.2)

The options for sovereign cloud, secure hosting, data storage and computing infrastructure have no price reference: C_o is the total amount entered by the user, usually from a quotation or an existing contract. Until an amount is entered, the option is shown as "missing amount" and stays out of D.

Additional costs

The three additional costs are percentages of the direct cost: travel applies to D; overhead and the administrative fee apply to D + V, and neither enters the other's base.

$$V = p_V\, D, \qquad O = p_O\, (D+V), \qquad T = p_T\, (D+V)$$

Cost	Default	Base	Source of the default (grade B)
Travel and in-person activities (fares, per diems, workshops, technical meetings)	25%	D	Global Fund: travel-related costs (TRC) are 24.7% of programme budgets in cycle 7, excluding health products (27.6% in cycle 6; 28.2% in cycle 5)
Overhead (indirect costs: premises, IT, HR, accounting, legal)	15%	D + V	United States, 2 CFR 200.414(f): up to 15% of direct costs without a negotiated rate (2024 revision)
Administrative fee (charged by whoever executes the funds)	7%	D + V	UNSDG policy for UN pooled funds: 7% of programme costs

The three questions

Each additional cost has a question that says whether it applies. Without an answer, the default in the table applies; the answer that excludes the cost sets its percentage to zero.

Question	Answer that excludes the cost	Effect
Will the plan have technical meetings, workshops or travel?	No	p_V = 0; overhead and the fee then apply to D only
Does the institution already cover indirect costs?	Yes, it already does	p_O = 0
Who will execute the funds?	The agency itself	p_T = 0; with a UN agency, 7%; with a support foundation, the user enters the contract fee (in Brazil, up to 15%, Decree 9,283/2018)

Any of the three percentages can be changed in the tool. For comparison, the European Union uses 7% indirect costs in ordinary grants and 25% in Horizon Europe, and UNESCO charges on average 8% in programme support costs.

Duration, exchange rate and totals

Governance has no phases: the plan has a single period of duration H (default 12 months, editable), and all costs are added up within it.

A one-off option counts all its reference months, even if they go beyond H (for example, a 12-month sandbox starting in month 4 of a 12-month plan). The implementation roadmap flags these cases in its calendar, but the cost is not cut.

The exchange rate x is the one entered in the case, if any; otherwise it is the ExchangeRate-API reference rate (open.er-api.com, USD base, 3 October 2026), rounded to 4 decimals so that the calculation can be checked by hand. Amounts in different currencies are never added together.

Each line has one of three statuses, and the total shows the first two separately:

Status	When	In the total
Entered	All the line's values were entered by the user	Yes
Estimated	At least one value comes from a reference (country earnings, reference team, default percentage)	Yes, marked as estimated
Missing amount	A quantity, price or months value is missing	No; shown as pending

The known cost is the sum of entered and estimated amounts. Values are calculated to the cent; the roadmap rounds totals to the thousand.

Reference team parameters

The 25 reference teams have 2 to 8 people; one-off options last 4 to 12 months and ongoing ones the whole plan. ISCO-08 profiles: G1 managers, G2 professionals, G3 technicians and associate professionals, G4 clerical support workers. The last column names the public case used to check the order of magnitude.

Cat.	Option	Months	People	Full-time equivalent	Team: role (profile) people × time share	Comparison
1.1	AI safety and risk classification	8	7	5.5	Coordination (G1) 1 × 0.5; Digital law and regulation specialist (G2) 2 × 1; AI and risk assessment specialist (G2) 1 × 1; Education policy specialist (G2) 1 × 0.5; Public consultation analyst (G3) 1 × 1; Administrative support (G4) 1 × 0.5	
1.1	Children's rights and protection	6	6	3.5	Coordination (G1) 1 × 0.5; Children's rights specialist (G2) 1 × 1; Child development and education specialist (G2) 1 × 0.5; AI specialist (G2) 1 × 0.5; Public consultation analyst (G3) 1 × 0.5; Administrative support (G4) 1 × 0.5	
1.1	Personal data protection and privacy	8	7	5.5	Coordination (G1) 1 × 0.5; Data protection law specialist (G2) 2 × 1; Education data specialist (G2) 1 × 1; Data protection officer (G2) 1 × 0.5; Public consultation analyst (G3) 1 × 1; Administrative support (G4) 1 × 0.5	Kenya data protection office
1.1	Algorithmic fairness, non-discrimination, transparency, explainability and accountability	9	8	6.5	Coordination (G1) 1 × 0.5; AI ethics and fairness specialist (G2) 1 × 1; Data scientist (G2) 1 × 1; AI explainability specialist (G2) 1 × 1; Fundamental rights specialist (G2) 1 × 1; Inclusive education specialist (G2) 1 × 0.5; Technical analyst (G3) 1 × 1; Administrative support (G4) 1 × 0.5	
1.1	Cybersecurity and incident reporting	6	6	4.5	Coordination (G1) 1 × 0.5; Information security specialist (G2) 2 × 1; Digital law specialist (G2) 1 × 0.5; Incident response analyst (G3) 1 × 1; Administrative support (G4) 1 × 0.5	
1.1	Intellectual property and copyright	5	5	2.5	Coordination (G1) 1 × 0.25; Intellectual property specialist (G2) 1 × 1; Educational resources specialist (G2) 1 × 0.5; Generative AI specialist (G2) 1 × 0.25; Administrative support (G4) 1 × 0.5	
1.2	Controlled testing environments	12	8	6.5	Coordination (G1) 1 × 1; AI specialist (G2) 1 × 1; Regulation specialist (G2) 1 × 1; Educational assessment specialist (G2) 1 × 0.5; IT infrastructure and security specialist (G2) 1 × 0.5; Pilot school liaison (G3) 1 × 1; Technical analyst (G3) 1 × 1; Administrative support (G4) 1 × 0.5	Norway sandbox
1.2	Regulatory supervision	ongoing	7	7	Coordination (G1) 1 × 1; AI supervision specialist (G2) 3 × 1; Administrative law specialist (G2) 1 × 1; Technical analyst (G3) 1 × 1; Administrative support (G4) 1 × 1	AI Act, COM(2021) 206
1.2	Risk assessments	6	4	2.25	AI risk assessment specialist (G2) 1 × 1; Data scientist (G2) 1 × 0.5; Data protection specialist (G2) 1 × 0.25; Technical analyst (G3) 1 × 0.5	Norway sandbox
1.3	Ethical AI guidelines for education	5	5	3	Coordination (G1) 1 × 0.25; AI ethics specialist (G2) 1 × 1; Education specialist (G2) 1 × 1; Technical writer (G3) 1 × 0.5; Administrative support (G4) 1 × 0.25	
1.3	AI procurement and vendor evaluation guidelines	5	6	3.5	Coordination (G1) 1 × 0.25; Public procurement specialist (G2) 1 × 1; AI specialist (G2) 1 × 1; Legal specialist (G2) 1 × 0.5; Technical analyst (G3) 1 × 0.5; Administrative support (G4) 1 × 0.25	

Cat.	Option	Months	People	Full-time equivalent	Team: role (profile) people × time share	Comparison
1.3	Age-appropriate AI use guidelines	4	5	2.5	Coordination (G1) 1 × 0.25; Child development specialist (G2) 1 × 1; Education specialist (G2) 1 × 0.5; AI specialist (G2) 1 × 0.5; Administrative support (G4) 1 × 0.25	
1.3	Teacher and student AI-use policies	4	6	2.75	Coordination (G1) 1 × 0.25; Education specialist (G2) 1 × 1; Assessment and academic integrity specialist (G2) 1 × 0.5; Legal specialist (G2) 1 × 0.25; Technical analyst (G3) 1 × 0.5; Administrative support (G4) 1 × 0.25	
1.3	AI risk management standards	5	5	3	Coordination (G1) 1 × 0.25; Risk management specialist (G2) 1 × 1; AI specialist (G2) 1 × 1; Technical analyst (G3) 1 × 0.5; Administrative support (G4) 1 × 0.25	
1.3	AI transparency and documentation standards	4	5	2.5	Coordination (G1) 1 × 0.25; AI specialist (G2) 1 × 1; Legal specialist (G2) 1 × 0.5; Technical analyst (G3) 1 × 0.5; Administrative support (G4) 1 × 0.25	
1.3	Data governance and security standards	5	5	3	Coordination (G1) 1 × 0.25; Data governance specialist (G2) 1 × 1; Information security specialist (G2) 1 × 1; Technical analyst (G3) 1 × 0.5; Administrative support (G4) 1 × 0.25	
1.3	AI quality assurance and evaluation guidelines	5	5	3	Coordination (G1) 1 × 0.25; Educational assessment specialist (G2) 1 × 1; AI specialist (G2) 1 × 1; Data analyst (G3) 1 × 0.5; Administrative support (G4) 1 × 0.25	
2.1	Data governance, protection and digital public infrastructure	ongoing	7	6.5	Coordination (G1) 1 × 1; Data governance specialist (G2) 2 × 1; Data and interoperability architect (G2) 1 × 1; Data analyst (G3) 2 × 1; Administrative support (G4) 1 × 0.5	Kenya data protection office
2.3	Open-source adoption and maintenance	ongoing	4	3.5	Technical lead (G1) 1 × 0.5; Developer (G2) 2 × 1; Support technician (G3) 1 × 1	
2.3	Technology transfer	8	6	3.75	Coordination (G1) 1 × 0.5; IT specialist (G2) 2 × 1; Contract manager (G2) 1 × 0.5; Technical trainer (G3) 1 × 0.5; Administrative support (G4) 1 × 0.25	
2.3	Local customisation	8	6	5	Coordination (G1) 1 × 0.5; Developer (G2) 2 × 1; Curriculum specialist (G2) 1 × 1; Language and translation specialist (G2) 1 × 0.5; Tester (G3) 1 × 1	
2.3	Vendor dependency reduction	4	4	2.25	Coordination (G1) 1 × 0.25; Public procurement specialist (G2) 1 × 1; IT architect (G2) 1 × 0.5; Legal specialist (G2) 1 × 0.5	
3.3	Governance committees	ongoing	2	1	Executive secretary (G2) 1 × 0.5; Administrative support (G4) 1 × 0.5	
3.3	Interministerial coordination	ongoing	3	2.5	Coordination (G1) 1 × 1; Public policy analyst (G2) 1 × 1; Administrative support (G4) 1 × 0.5	
3.3	Institutional governance structures	ongoing	8	8	Coordination (G1) 1 × 1; AI specialist (G2) 1 × 1; Data specialist (G2) 1 × 1; Legal specialist (G2) 1 × 1; Education specialist (G2) 1 × 1; Technical analyst (G3) 2 × 1; Administrative support (G4) 1 × 1	Kenya data protection office

Team composition, time shares and months are tool conventions (grade C), designed the way a coordinator would build the team: who leads, the specialists for each area and support staff. The supervision unit sits at the lower end of the range the European Commission estimated for the AI Act (1 to 25 full-time staff per country); the sandbox and the risk assessments follow the pace of Norway's sandbox (4 projects a year, 3 to 6 months each).

Evidence grades and limitations

Earnings and employer contributions have public sources; team composition, the generic US$ references and the calendar are tool conventions and are the parameters that most need calibration.

Parameter	Grade	Source
Earnings by profile, country with data	A	ILOSTAT, by country and year
Earnings by profile, country without data	B	Regional or income-group median
Employer contributions	A	SSA and ISSA, table 4
Travel, overhead and fee percentages	B	Global Fund, 2 CFR 200.414, UNSDG
Exchange rate	A	ExchangeRate-API, or entered in the case
Team composition, time shares and months	C	Tool convention, compared with public cases
Annual loaded cost of US$36,000 and US$45,000	C	Tool convention (v13), illustrative
Training cost of US$12 per hour	C	Tool convention (v13), illustrative
Start month and typical timeframes in the roadmap	C	Tool convention

The model has six known limitations:

1. ILOSTAT average earnings cover all employees in the occupational group, in the public and private sectors; AI, digital law and data specialists tend to earn above this average, so the model tends to understate the cost of these roles.
2. Earnings data range from 2015 to 2025 depending on the country and are not adjusted for inflation to the plan year.
3. The employer contribution does not include benefits paid directly by the employer, such as a 13th salary, paid leave, health plans or supplementary pensions.
4. A person who holds the same role in more than one option (for example, the coordinator) is counted once in each option; the model does not net out this overlap.
5. The cost of existing civil servants is counted as if it were new; the plan shows the economic cost of the work, not the additional cash outlay.
6. There is no uncertainty interval: each line has a point value. The plausible range can be explored by changing the parameters in the tool.

Worked example: Brazil test case

In the test case (Brazil, Brazilian reais, 12 months, 12 cost lines), the known cost is R$4,964,082.54: R$3,255,136.09 in options and R$1,708,946.45 in additional costs, with two options still without an amount.

Monthly cost by profile

Brazil has its own ILOSTAT data (Continuous PNAD, 2025), in reais, and an employer contribution of 21% (SSA and ISSA, the Americas, 2019); the contribution factor is 1.21.

Profile	Average monthly earnings, w_g (R$)	Monthly cost with contributions, c_g (R$)
G1 Managers	8,000.59	9,680.71
G2 Professionals	6,551.79	7,927.67
G3 Technicians and associate professionals	4,332.05	5,241.78
G4 Clerical support workers	2,762.83	3,343.02

One option step by step

The option "AI safety and risk classification" has 8 months and six roles. The team's monthly cost is R$39,500.49, and the option costs R$316,003.92.

$$\begin{aligned}\text{monthly cost} &= 1 \times 0.5 \times 9{,}680.71 + 2 \times 1 \times 7{,}927.67 + 1 \times 1 \times 7{,}927.67 \\ &\quad + 1 \times 0.5 \times 7{,}927.67 + 1 \times 1 \times 5{,}241.78 + 1 \times 0.5 \times 3{,}343.02 = 39{,}500.49 \\ C_o &= 8 \times 39{,}500.49 = 316{,}003.92\end{aligned}$$

All lines in the case

Cat.	Option	Monthly cost (R$)	Months	Cost (R$)
1.1	AI safety and risk classification	39,500.49	8	316,003.92
1.1	Algorithmic fairness, non-discrimination, transparency, explainability and accountability	47,428.16	9	426,853.44
1.2	Controlled testing environments	45,618.79	12	547,425.48
1.2	Regulatory supervision	49,976.19	12 (ongoing)	599,714.28
1.3	AI procurement and vendor evaluation guidelines	25,696.00	5	128,479.99
1.3	Teacher and student AI-use policies	19,750.24	4	79,000.98
2.1	Data governance, protection and digital public infrastructure	45,618.79	12 (ongoing)	547,425.48
2.2	Secure hosting			missing amount
2.3	Open-source adoption and maintenance	25,937.48	12 (ongoing)	311,249.70
3.2	Training and professional development			missing amount
3.3	Governance committees	5,635.35	12 (ongoing)	67,624.14

Cat.	Option	Monthly cost (R$)	Months	Cost (R$)
3.3	Interministerial coordination	19,279.89	12 (ongoing)	231,358.68
	Direct cost, D			3,255,136.09
	Travel, 25% × D			813,784.02
	Overhead, 15% × (D + V)			610,338.02
	Administrative fee, 7% × (D + V)			284,824.41
	Known cost, C			4,964,082.54

With the three defaults, the additional costs add 52.5% to the direct cost. If the plan has no travel, the institution already covers indirect costs and the funds are executed by a foundation charging 7%, the additional costs fall to R$227,859.53 (7% × D) and the known cost to R$3,482,995.62.

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